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Dilemmas and challenges in forest conservation and development interventions: Case of Northwest Pakistan

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ABSTRACT

Conservation and development projects are key instruments of international development agencies for natural resource conservation and rural development. However, despite some success stories, conservation of natural resources (particularly forests) in conjunction with socio-economic development – which is the precondition for sustainable development – remained a challenge in most of the developing countries. In Pakistan, too, natural forests in the Northwestern highlands continue to be depleted in spite of numerous interventions, by the international donors, to conserve the remaining forests. This paper uses a sustainable development perspective, and attempts to study the quest by the forest conservation and development interventions – initiated by the overseas development aid – regarding operationalisation of sustainable development as conceived by the projects' implementing agencies and thereby comparing it with local implementation context in terms of perceived impact/usefulness, and participation of stakeholders in the projects. This paper argues that, without considering socio-economic realities at micro (village) level, one cannot envisage the success of conservation and development interventions by considering only structural and meso (national/regional) levels. Forestry should be seen in a multi-stakeholder scenario where various actors have different claims and entitlements over forest resources. Major challenge for international development donors is to ensure a balance of power between stakeholders.

1. Introduction

Widespread deforestation in many developing countries impelled the international development community to make huge investments in the form of conservation and development projects during the last two decades (FAO, 2007; Siry et al., 2005). In most of the countries such conservation efforts led to the emergence of participatory forest management policies for effective conservation and sustainable governance of forest resources. Consequently institutional changes, decentralization and joint forest management paradigms have become major trends in the forestry sector of most of the developing countries (Larson and Ribot, 2004; Lescuyer et al., 2001; Sunderlin et al., 2005). However transformation of forest management to a multiple stakeholder community-based approach remained a major challenge, and in many developing countries of Asia and Africa deforestation continues at high rate (FAO, 2007) though massive support has been provided by the international development agencies. Institutional reforms in the forest sector have long-term impacts so there is a need to carry out major evaluations of interventions and policies (Nilsson, 2005). The linkages, complementarities and trade-offs between development and environmental objectives of the conservation & developments interventions in the forestry sector has been studied intensely and many researchers believed that most of the interventions were futile in achieving their objectives (Marcus, 2001; Sunderlin et al., 2005) but still many development researchers and practitioners are concerned to explore the causes of ineffectiveness of such interventions by using alternative perspectives (Gray and Moseley, 2005; Grimble and Wellard, 1997).

In the mountainous regions of Khyber Pakhtunkhwa (KP) province of Pakistan (formerly known as North West Frontier Province, NWFP) a series of conservation and development interventions had been initiated during the last two decades in response to massive deforestation in this part of the world (Bhatia, 2000). Provincial Forest Department, being the legitimate custodian of natural forests, remained nucleus of such activities (Ahmed and Mahmood, 1998). The intention of most of the interventions was to mitigate the problems caused by deforestation by involving and motivating local communities to plant trees and protect the watershed areas. Most of these interventions were supported by the international development (donor) agencies. Major policy shifts were also made to provide legal cover to the new (participatory) paradigm of forest governance (see for example Shahbaz et al., 2007; Simorangkir, 2006). With the exception of few projects, most of the interventions couldn't achieve

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their intended objectives and thus despite decade long conservation efforts and optimistic developments – including a conducive policy climate – the deforestation in the highlands of Northwest Pakistan continued (FAO, 2007; Shahbaz and Suleri, 2009). The issues of forest depletion in the KP and the impact of conservation projects has remained an emerging area of research during the recent years (see for example Ali et al., 2006; Simorangkir, 2006; Steimann, 2003).

Provincial Forest Department of KP is the mandated caretaker of the forests (as per constitution of Pakistan) and thus is a major stakeholder but there also exists a wide range of actors who do have a stake in the use of forest resources. The key stakeholders include; forest land owners/right-holders, landless forest users including graziers (known as *Gujjars*), the state (provincial forest department and federal government), private sector, civil society including community based organizations, traditional institutions, timber smugglers etc. (Ahmed and Mahmood, 1998; Rome, 2005; Steimann, 2003; Shahbaz, 2009). These stakeholders have different rights and claims in the forest. For instance, in many regions of the province, where traditional rights had existed regarding the access and benefit sharing of forest resources, the traditional forest owners and right holders have never accepted the state control over forests as they still claim ownership of the forests and perceive state as their competitor in the utilization of forest resources (Rome, 2005). However in most parts of Hazara division of the province the property rights have been defined relatively clearly. Nevertheless the local communities are not homogeneous and there exist various groups of actors within one community (see also Steimann, 2005) and these groups not only have internal conflicts but also many groups challenge legitimacy of the state (Shahbaz et al., 2008).

The forestry sector of Northwest Pakistan is, therefore, an interesting case to analyze the efforts and intentions (in the form of conservation and development interventions) of the international development community for sustainable forest governance with reference to the local perspective. This article presents empirical evidences from a research project that examined the intentions/activities of conservation and development projects in the Khyber Pakhtunkhwa province of Pakistan and comparing these with the local perceptions (at village level).

1.1. Theoretical framework

Conservation and development projects are major instruments of natural resource protection and local development, and the study of nexus between natural resource conservation and socio-economic development has remained an overarching theme of development research (Gray and Moseley, 2005; Marcus, 2001; Sunderlin et al., 2005). Precise theorisation on the relationship between forest conservation and socio-economic development is contested (Wunder, 2001), however the need for holistic understanding of linkages and trade-offs between environmental and development objectives has been emphasized by the researchers (Grimble and Wellard, 1997; Mahanty, 2002). Development practitioners and researchers agree that degradation of natural resources (particularly forests) affects the livelihoods of those who are fully or partly dependent on such resources (Sunderlin et al., 2005); therefore the natural resource conservation (and management) is being used as a tool of rural development and vice-versa. Since the Brundtland report (WCED, 1987) and the UN Summit in Rio de Janeiro in 1992, the notion of sustainable development has become the guiding theme in most of the environmental literature. The rise of sustainable development concept triggered the idea of development vis-à-vis conservation (Gray and Moseley, 2005). Integrated conservation and development projects have evolved as part of global agenda to achieve the goals of sustainable development (Mahanty, 2002) and participation of local communities in the governance of natural resource is a significant feature of such internationally supported programs throughout the world (Larson and Ribot, 2004). Nevertheless, various researchers have indicated the constraints in the implementation of global agenda of forest conservation vis-à-vis sustainable development (Leach et al., 1999; Nilsson, 2005; Ribot, 2004), and strong disconnections are observed between the intentions of the global development agencies and the local implementation context (Marcus, 2001). It has been argued that governance of the natural resources (particularly forests) is influenced by a complex interplay of various actors and institutions operating at multiple scales — from micro (village) to macro or national and global level (Leach et al., 1999). For example an environmental degradation phenomenon (such as deforestation) can not only have local and regional contexts but also global dimensions, because various actors and processes work at different levels (Beratan, 2007; Leach et al., 1999).

This paper considers the implications of these insights on sustainable development perspective of conservation and development programs and aims to add to the emerging discussions on this area of research by drawing on the case study of the forestry sector in Khyber Pakhtnkhwa province of Pakistan. By using a sustainable development perspective, the paper attempts to study the quest by the forest conservation and development interventions – initiated by the overseas development aid – regarding operationalisation of sustainable development and thereby comparing it with local implementation context in terms of perceived impact/usefulness, and participation of stakeholders in the projects.

2. Methodology

Khyber Pakhtunkhwa (former North West Frontier Province) was purposely chosen because it has maximum forest cover than other provinces and territories of Pakistan. About 40% of the country's natural forests are located in Khyber Pakhtunkhwa province and almost 17% of the total land area of the province is covered by forests (Ahmed and Mahmood, 1998). Within the province most of the natural forests are located in the mountainous districts viz. Chitral, Upper Dir, Lower Dir, Swat, Shangla, Malakand, Buner, Kohistan, Battagram, Mansehra, Abbottabad and Haripur The province covers an area of 10.17 million hectares with a population of approximately 15 millions. It is located on both banks of the river Indus and is bounded by the Himalaya, Korakoram and Hindukush Mountains in the north and Afghanistan in the north-west. The major apprehensions of the rural areas of the province are poor infrastructure, poor health and educational facilities, limited access to economic opportunities, high illiteracy natural resource degradation, geographical isolation, natural disasters (earthquakes, floods), low HDI etc. (Halle et al., 2004; Steimann, 2003; Shahbaz, 2009). For these reasons overall incidence of poverty in the rural Khyber Pakhtunkhwa is substantially higher than that for the country as a whole (Halle et al., 2004).

The main objective of this research project was to identify and analyze the efforts and intentions of the international development community for sustainable forest governance (in the form of conservation and development interventions) and comparing them with the local context. Therefore all of the completed (or ongoing) conservation and development projects/interventions related to forestry in the highland districts of the Khyber Pakhtunkhwa province during the years 1997–2008 were taken as population of the study. In the first step of data collection an inventory was prepared by getting information from Planning and Development Department of the provincial government, websites of the governmental and non-governmental organization, and international donor agencies. Information was reconfirmed through personal visits in the study area and telephonic calls to different departments and organizations. A structured questionnaire was used as an instrument to collect information regarding intentions and activities of each project/intervention in the perspective of sustainable development and forest management. After examining the validity and reliability of the instrument (Bernard, 2000), the questionnaire was sent by mail to the responsible persons (for example project director, senior forest officials, senior representative of donor agency etc.) of the respective implementing organization of the intervention. To ensure 100% response rate, face to face and telephonic interviews were also conducted.

The second step of the research was conducted with the help of M.Sc. students (Wattoo, 2009; Kiran, 2009). The objective of this step was to identify and analyze the perceptions of local communities regarding usefulness, involvement and participation of stakeholders in the forest interventions. There are two distinct geographical regions in the province where natural forests are located i.e., areas in East of river Indus (Hazara division) and areas in the West of River Indus (Malakand division). The village survey was not possible in the Malakand division due to ongoing insurgency and war on terror, therefore only those districts were selected which are comparatively "safe" and located at the east of river Indus. These districts include, Haripur, Abbottabad, Mansehra, Battagram, Kohistan and Chitral. From these districts 10% (Six projects) were randomly selected from the total of 57 interventions. These projects were operational in Haripur, Abbottabad and Mansehra districts. By applying multistage sampling technique, one village was selected from each project and 40 respondents from each village were selected randomly. Thus in total 240 respondents from six villages were interviewed with the help of a semi-structured questionnaire. Qualitative data were also collected through key-informant interviews and focus group discussions with the help of interview guide (Bernard, 2000; Patton, 2002).

3. Results

This paper uses a sustainable development perspective of the conservation and development projects and compares the intentions of the projects conceived at by the implementing agencies of the projects at national as well as international level, with the local perceptions at village level. This section therefore presents the valuation and actions for the attainment of sustainable development as conceived by the projects' implementing agencies and perception of the villagers (respondents) regarding their involvement and usefulness of the interventions.

3.1. Intentions and actions of the interventions

The importance of forests in economic, environmental and socio-cultural domains of sustainable development is not a contested issue as forests are important not only for local population but also for the overall sustainable development of a country (Salam et al., 2005). In this milieu the relative priorities of the conservation and development interventions on three domains of sustainable development viz. economic, socio-cultural and environmental sustainability were recorded from the projects' implementing agencies. The responsible persons (project director, senior forestry official etc.) from implementing agencies were asked to distribute a total of 10 points among the three dimensions in such a way that sum of all three values should be equal to 10 where higher number indicates higher priority. In this way different priorities of projects can be judged more effectively because they have to maintain a balance between different dimensions of sustainable development. The results reveal that the environmental sustainability with highest mean value (4.18) was top priority followed by economic sustainability with 3.54 and socio-cultural sustainability (mean value 2.28).

Priorities (and actions) of different projects to ensure environmental sustainability were also measured on an 11-point likert scale (with 0 indicating no priority and 10 representing highest priority). Unsurprisingly most of the projects prioritized forest protection followed by integrated natural resource management, new plantation and forest regeneration for ensuring environmental sustainability in their respective areas. Climate moderation aspect was given less than average importance.

The actions of different projects to ensure economic sustainability in their respective regions were recorded on an eleven-point likert scale. Trainings of the villagers on forest related issues remained the top prioritized action followed by income generation (through forest resources) and employment. Income related support for small business and micro credit were comparatively lesser prioritized areas for ensuring economic sustainability.

The results regarding socio-cultural dimension indicate that the empowerment of local communities (through their participation in the forest related issues) was top most priority of the conservation and development projects. Support for traditional institutions was ranked second followed by gender mainstreaming.

3.2. Local realities

3.2.1. Participation in project meetings & activities

Participation of local users in projects' activities is a central component of most of the conservation interventions (Section 3.1). The respondents were asked to rate the participation of their households and their fellow villagers in project's meetings & other related activities on a scale with 0 indicating no participation and 5 representing very high participation. The results revealed that the extent of participation of respondent's households and other peoples of the village in project meetings was perceived as low and very low respectively. Some of the qualitative remarks further enlightened the situation.

A counselor (local elected leader) explained, "The project staff meets only with the Forest Department and some influential persons. It is natural because they have to run their project and without the support of Forest Department they can't succeed". A farmer commented, "(........) participation in meetings? I don't know about any meeting held in our village so there is no question of participation in the meeting. Yes they have done some plantation and constructed a track which is good for us but we were not consulted for this".

3.2.2. Participation in decision-making

Involvement of stakeholders in forest management has been the major thrust of forest related conservation and development interventions (see Section 3.1). In other words, the role of Forest Department should gradually be reduced and the local stakeholders should have been given more responsibility in the forestry issues. In order to extricate reality the respondents were asked to rate the role of different institutes/departments regarding forest related issues (particularly on planting and harvesting of trees) on a six-point likert scale (not at all = 0, very low = 1, ..., very high = 5) and their perception in tabular form are given in Table 1.

The respondents perceived that the provincial Forest Department, with an overall highest mean value of 4.66, possesses very high role in decision making on forest management related issues specifically on the harvesting of trees. The qualitative data revealed that this perception was because of the fact that the local forest owners have to formally request the Forest Department to obtain permission to cut even a single pine tree for domestic (subsistence) use and they have

Table 1
Perceived role in decision making regarding forest management (particularly harvesting) at local level.

Role in decision-making	Mean	Standard deviation	Ranked order
Your household's role	0.12	0.33	5
Local communities role	0.29	0.57	4
Forest Department's role	4.66	0.41	1
NGOs role	2.54	0.67	2
Local *Jirga's* role	3.03	0.79	3

to get the approval from a hierarchy of forest officials (see Section 4 for details). Therefore, most of the respondents perceived that the Forest Department has highest role in decision-making at local level. The *jirgas* (informal assembly of elders) was ranked at second place and the NGOs were perceived as below than average regarding their roles in forest related issues. The role of local communities was perceived to be very low in decision-making regarding harvesting of trees for domestic use.

The qualitative interviews also revealed that, despite the claim (and intention) of most of the projects regarding community empowerment (see Section 3.1), the respondents still perceive the Forest Department as most influential actor. For instance a farmer commented "(.......) they (the Forest Department) are very powerful people and they decide when to harvest trees. We are totally dependent on them for wood". Another person narrated "*Jirga* is composed of rich and big forest owners, and the personnel from Forest Department only consult them some times but they (the Forest Department) decide on their own way how to manage forests".

The above quotes clearly indicate the authoritative behavior of the state officials toward local forest users. In a recent study, Wattoo et al. (2010) endorsed that the interaction of forest department and local communities is very low and involvement of local people in decision making process in the context of forest related interventions in the Northwest Pakistan is also very low.

3.2.3. Perceived usefulness of the projects

Respondents were asked to rate the projects' usefulness on economic, environmental and socio-cultural dimensions of sustainable development. Their responses were rated on a six-point likert scale (not at all = 0, very low = 1, low = 2, ..., very high = 5).

The data presented in Table 2 indicate the economic, environmental and socio-cultural usefulness of the project as perceived by the respondents. The overall usefulness of the projects in terms of three dimensions of sustainable development was rated as very low. The respondents perceived that the interventions didn't bring any significant changes in the economic or environmental status of the area. Similarly the projects' benefits in terms of socio-cultural aspect were also perceived as very low. Nevertheless, with highest overall mean value 1.22 respondents perceived the environmental usefulness of the project just more than very low. Some of the qualitative remarks further depicted the picture.

"(........) we need food for our children, and money to buy food and to send our children to schools. The projects have very little concerns to meet our economic needs". "(...........) no doubt forests are good for us but if these forests cannot provide us income then these are useless for us. They (projects and Forest Department) are concerned with trees and they have no worries about our economic needs. They arranged some training workshops but very few people were invited". "(.......) we are dependent on governmental officials for wood therefore we see no benefits of our involvement in project activities."

The above qualitative data clearly illustrate that the villagers are more concerned with satisfying their economic needs. They are dependent on the Forest Department for obtaining timber for domestic purpose; however the project staff and the Forest Department are more concerned with the forest conservation.

Table 2
Perceived economic, environmental and socio-cultural usefulness of the project.

Dimensions of sustainable development	Mean	Standard deviation	Rank order
Economic usefulness	0.93	1.02	3
Environmental usefulness	1.22	1.36	1
Socio-cultural usefulness	1.17	1.4	2

4. Discussion: dilemmas and challenges

The focus of the paper is an analysis of forest related interventions by appraising the intentions and actions conceived by the projects' implementing agencies (mostly working at macro i.e., national and international levels) and then digging out the reality by going down at the micro (village) level and exploring local perceptions regarding their participation in the conservation projects, involvement in the decision making (particularly harvesting of trees for domestic use) and socio-economic and environmental usefulness of projects.

The qualitative and quantitative data given in the above section expose some of the dilemmas in conservation projects in KP. On one hand, main thrust of most of the interventions is environmental sustainability through resource conservation and involvement of the local stakeholders or community empowerment (Section 3.1) but on the other hand, the data collected at the village level reveal that, participation was not really done at micro (village) level. Majority of the respondents reported that they didn't participate in the project activities and perceived Forest Department as most powerful actor and they (the local people) are the least influential regarding decision making in resource utilization (particularly harvesting of trees for domestic use) which is quite contrary to the thrust of community empowerment by development interventions during the past two decades (see Section 3.1). Most of the respondents informed that they need fuel–wood and construction timber for their household needs. The local reality is that the residents (forest land owners/right-holders) have to obtain formal permit from Forest Department for timber for subsistence purpose but the permit procedure is rather complicated. The application of the right holders has to be routed through a hierarchy of officials and a lot of time is required in the process therefore usually the dwellers exercise informal means such as illegal cutting of trees or giving bribes to the local forest guards. In the context of participatory forest management paradigm the forest owners demanded simplification in the permit procedure and they need more empowerment in harvesting of trees for domestic use. However forest conservation is the top most priority of the Forest Department and there are little concerns about the local subsistence needs. Some of the relevant excerpts from qualitative interviews are given as under;

A forest landowner informed "My house was damaged due to heavy snowfall during the last winter and I needed timber to repair the roof. I had to request the forest guard who demanded money and then he signed my application." Another angry respondent commented, ".......... the project people have established a village committee but this committee has no powers and therefore is useless for us. We still look at the forest officials for timber for our own use".

The authoritative behavior of the forestry officials that has its roots in colonial period (Rome, 2005) is one of the major factors hindering the effectiveness of participatory approach. The Forest Department wants to maintain its authority over forests and the conservation-development interventions have to involve Forest Department as key player – most of the times as major implementer of the project – but in such situation local stakeholders feel themselves uncomfortable because of the attitude of the state officials and the dependence of people on them for forest related issues (see also Geiser and Steimann, 2004; Shahbaz et al., 2008). Previous researchers have corroborated that the staff of the Forest Department (particularly lower field staff) resisted the participatory approach because they perceived that their authority is threatened (Ali et al., 2006, 2007; Steimann, 2003, 2005).

The other side of the picture revealed that the forests (particularly in the remote areas) are heavily overexploited by influential timber smugglers and for an honest forest officer it becomes pretty difficult to catch the offenders. For example a senior forest officer told "………. whenever we catch an illegal logger, my telephone and personal mobile phone start ringing with the calls from the political persons who want the release of the offenders". The Forest Department in the province has inadequate resources to deal with the illegal loggers. Low salaries, insufficient transport facilities and shortage of man power are also limiting factors in this regard (see also Geiser and Steimann, 2004).

Qualitative interviews revealed that illegal timber harvesting is the major reason of deforestation in the area. There exists a network of powerful elites (locally called as timber mafia) who are involved in illegal timber harvesting and smuggling. The timber mafia is highly organized and have strong linkages with the state officials. Nevertheless, the forest officials blame the influential persons for protecting the timber smugglers and there is also lack of incentives and facilities for the honest forest officials. Participatory forest management is considered as an effective strategy to deal with the illegal logging (Lescuyer et al., 2001); however this does not happen in the case of KP because for the local stakeholder groups participation didn't happen in reality (see Section 3.2).

The respondents also see very little usefulness of the intervention in terms of economic, environmental and socio-cultural aspects of sustainable development (Table 2). The economic concerns are top priority for the local dwellers, but the results of this study indicate that the respondents see no direct benefit of participation in the project activities. Although most of the projects have option of human resource development by providing different types of trainings but for the local communities such trainings were not of significant use because very few people participated in these trainings (compare Sections 3.1 and 3.2.1). The dilemma with most of the conservation interventions in Northwest Pakistan has been the absence of attention to economics aspects and a focus on conservation approach. Marcus (2001, 396) narrates, "The successful achievement of conservation goals thus ultimately rests on the ability of conservation and political advocates to provide the economic improvements the people want."

The lack of attention to human dimension aspects and a focus on conservation is another dilemma with most of the forest conservation efforts in KP and therefore disconnections are observed between the intentions of the development agencies regarding operationalisation of global agenda of sustainable development and the local implementation context. There are always trade-offs between socio-economic development and environmental conservation (Grimble and Wellard, 1997) and such development-environment nexus needs to be understood by an appropriate integration of disciplinary and interdisciplinary research (Gray and Moseley, 2005) as well as with reference to the interest groups and actors i.e. stakeholders.

There is a wide range of stakeholders who can influence or can be influenced by any forest related intervention in the Northwest Pakistan. These include local communities, civil society organizations, religious and traditional institutions, private sector, the state (federal and provincial ministries), global development agencies etc. These actors have different rights and claims in the forest resources (Rome, 2005; Steimann, 2003). In the context of participatory forest management the notion of "local communities" or "local people" has been used quite extensively and is regarded as the most important interest group, but recent research studies in the forest areas of Northwest Pakistan indicate that the local communities are not homogenous rather they are highly stratified entities (Simorangkir, 2006; Wattoo et al., 2010). Some of the groups within "local communities" in the province as identified by the previous researchers include forest owners/right-holders, non right-holder/landless forest users, and graziers (*Gujjars*). The forest owners are either the owners of *guzara* (subsistence) forests or have rights in the protected forests. The state has declared the forests as "state forests" but still pays royalty to the right-holders (in Malakand division) and thus the tribes having traditional ownership rights of forest lands still perceive themselves as owner of the forest but the forest laws restrict them in utilizing the forest resources. In Hazara division of the province, owners of the forest land can utilize forest resources for their subsistence/domestic use only after obtaining formal permission from the Forest Department. Non-owners and landless residents have to seek permission from owners/right holders for utilization of the forest resources. *Gujjar* tribe is a specific group of non-right holders in forests. This tribe is generally landless and relies on livestock to earn their living. Their access to the forest use is regulated through traditional norms and rules. Most of the *Gujars* families are nomads but some have permanently settled in certain areas.

The dilemma regarding the use of forest resources is that the state (Forest Department) considers forests as state property and expects that the "local communities" would follow the regulations of the Forest Department; but at the same time the traditional norms (known as *Rivaj*) prevail strongly in the rural areas of the province (Rome, 2005) and based on these traditional regulations the right-holders perceive themselves as forest owners and consider the state as an outsider enemy being in competition with the forest owners. Hence they resist any intervention that tries to restrict or regulate the forest uses (for details see Shahbaz et al., 2008; Geiser, 2000).

The number of interventions, projects, and policies in the forestry sector – no doubt – indicates the intensity of efforts and commitment made by the international community but the sustainable governance of the forests cannot and should not be measured from the number of such interventions rather ground level realities need to be extricated and dealt with accordingly. It is recommended that the local realities should be carefully analyzed because without clear understanding of realities at micro (village) level, one cannot think of the success of conservation and development interventions by considering only structural and meso (national/regional) levels. The economic concerns including subsistence needs of local communities and incentives for forest officials should be considered along with forest conservation. Forestry should be seen in a multi-stakeholder scenario where various actors have different claims and entitlements over forest resources and where the local communities are also heterogeneous and divided into different groups with different power relations. Major challenge for international development community and donors is to ensure a balance of power between state and the local stakeholders.

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